



IS231: Web Technology Web Architecture and HTML



By: Neamat El Tazi

References

Books:

- **Web Application Architecture: Principles, protocols and practices By Leon Shklar and Richard Rosen (Chapter 2)**
- **HTML Slides**
 - **Web Application Technologies and Django Coursera Course by University of Michigan. Taught by Professor Charles Russell**

Web Application Technologies

Client

Server

Browser

HTML
CSS
DOM
JavaScript
jQuery

Web Server

Django / Flask
Sqlite3 / MySQL

The First Page

If you like, you can switch to the [Second Page](#).

<http://mywebsite.com/page1.html>

Web Server

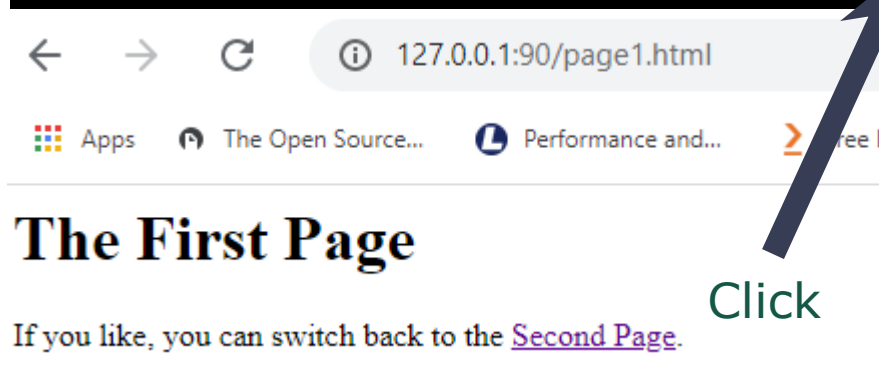
Request

GET
<http://mywebsite.com/page2.html>

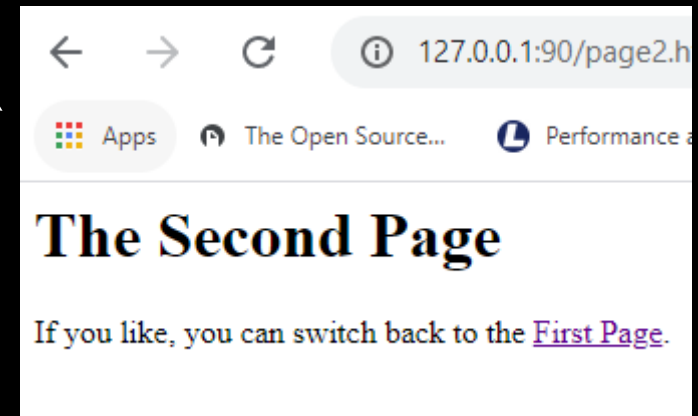
Response

```
<h1>The Second  
Page</h1>  
<p>If you like, you can  
switch back to the <a  
href="page1.html">First  
Page</a>.</p>
```

Browser



Parse/
Render



TCP Port Numbers

- ▶ A port is an application-specific or process-specific software communications endpoint
- ▶ It allows multiple networked applications to coexist on the same server
- ▶ There is a list of well-known TCP port numbers

http://en.wikipedia.org/wiki/TCP_and_UDP_port

Well-known ports

Incoming E-Mail

25

Web Server

80

Web Server

443

Personal Mail Box

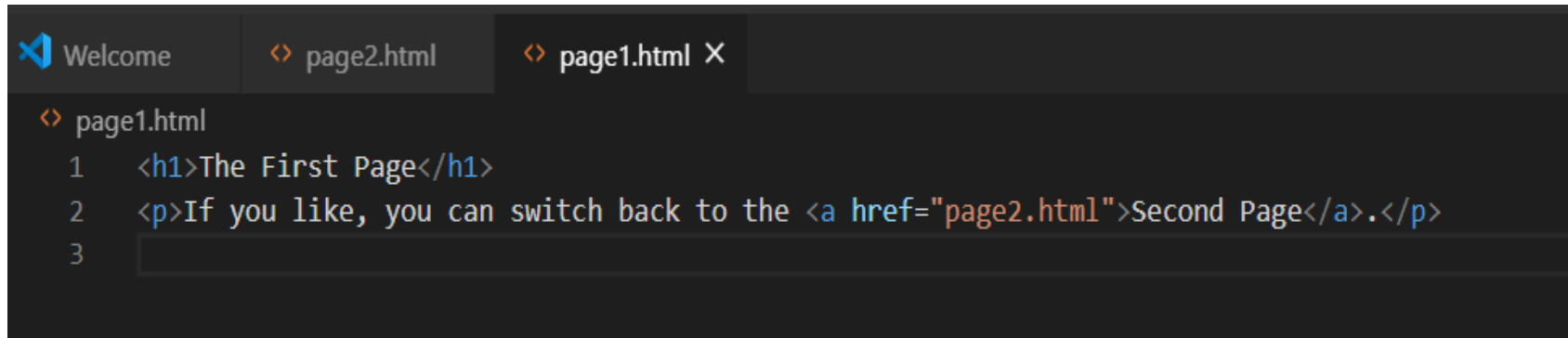
109

Personal Mail Box

110

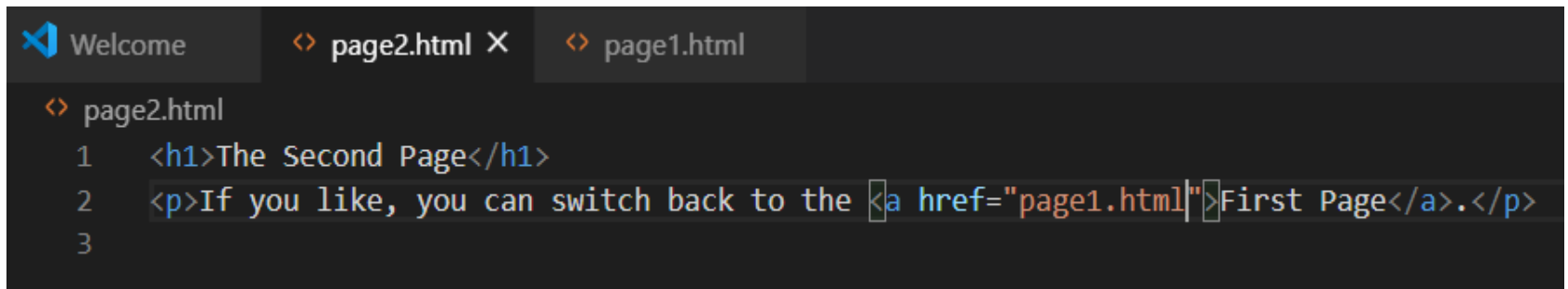
Demo

- ▶ Install python (using pip or Anaconda)
- ▶ Create a folder with two files (page1.html, page2.html)



A screenshot of the Visual Studio Code editor interface. The top tab bar shows three tabs: 'Welcome' (with a blue icon), 'page2.html' (with an orange icon), and 'page1.html' (with an orange icon and a close button 'X'). The 'page1.html' tab is active. The editor area shows the following HTML code:

```
<?xml version="1.0" encoding="utf-8" ?>
<html>
1  <h1>The First Page</h1>
2  <p>If you like, you can switch back to the <a href="page2.html">Second Page</a>.</p>
3
```



A screenshot of the Visual Studio Code editor interface. The top tab bar shows three tabs: 'Welcome' (with a blue icon), 'page2.html' (with an orange icon and a close button 'X'), and 'page1.html' (with an orange icon). The 'page2.html' tab is active. The editor area shows the following HTML code:

```
<?xml version="1.0" encoding="utf-8" ?>
1  <h1>The Second Page</h1>
2  <p>If you like, you can switch back to the <a href="page1.html">First Page</a>.</p>
3
```

Demo

- ▶ Run SimpleHTTPServer using python on port 90 (on the command prompt) :
 - ▶ `python -m SimpleHTTPServer 90` (If you are using Python 2)
 - ▶ `python -m http.server 90` (If you are using Python3)
- ▶ Go to your browser and go to the URL:
 - ▶ `127.0.0.1:90/page1.html`
 - ▶ Switch back and forth between page1 and page 2 using the links in the HTML pages
 - ▶ Check the output in the command prompt.

```
Serving HTTP on 0.0.0.0 port 90 ...  
127.0.0.1 - - [02/Nov/2020 08:17:56] "GET /page2.html HTTP/1.1" 200 -  
127.0.0.1 - - [02/Nov/2020 08:17:57] "GET /page1.html HTTP/1.1" 200 -  
127.0.0.1 - - [02/Nov/2020 08:18:00] "GET /page1.html HTTP/1.1" 200 -
```

Python 2

```
::ffff:127.0.0.1 - - [25/Feb/2023 13:41:08] "GET /page1.html HTTP/1.1" 200 -  
::ffff:127.0.0.1 - - [25/Feb/2023 13:41:10] "GET /page2.html HTTP/1.1" 200 -
```

Python3

HTML (Hyper Text Markup Language)

A way of marking up text to indicate that some text is different than other text

We “**tag**” portions of the text to communicate meaning

<p>You can add a style like
bold to some
text by enclosing it in the
appropriate
tag.</p>

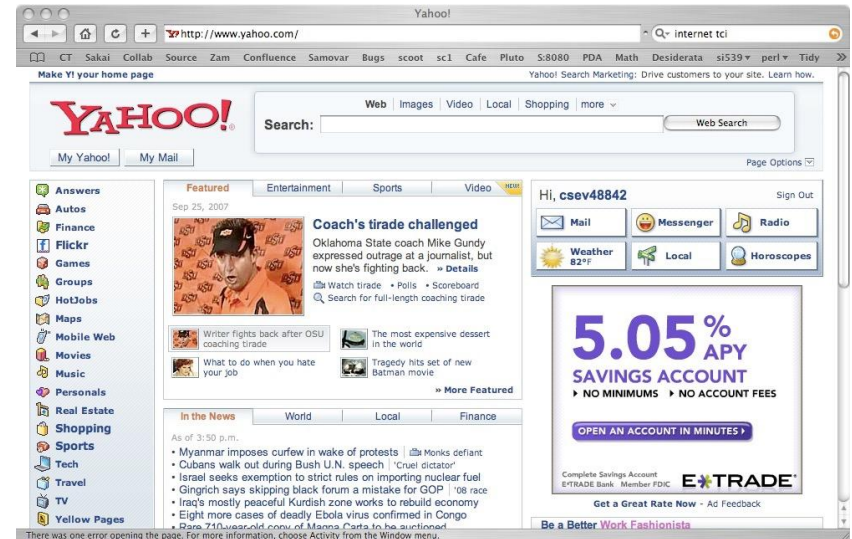
You can add a style like **bold** to some text by enclosing it in the *appropriate* tag.

Evolving HTML



- [Arts and Humanities](#) - [Architecture](#), [Photography](#), [Literature](#)...
- [Business and Economy \[Xtra!\]](#) - [Companies](#), [Investments](#), [Classifieds](#)...
- [Computers and Internet \[Xtra!\]](#) - [Internet](#), [WWW](#), [Software](#), [Multimedia](#)...
- [Education](#) - [Universities](#), [K-12](#), [College Entrance](#)...
- [Entertainment \[Xtra!\]](#) - [Cool Links](#), [Movies](#), [Music](#), [Humor](#)...
- [Government](#) - [Politics \[Xtra!\]](#), [Agencies](#), [Law](#), [Military](#)...
- [Health \[Xtra!\]](#) - [Medicine](#), [Drugs](#), [Diseases](#), [Fitness](#)...
- [News and Media \[Xtra!\]](#) - [Current Events](#), [Magazines](#), [TV](#), [Newspapers](#)...
- [Recreation and Sports \[Xtra!\]](#) - [Sports](#), [Games](#), [Travel](#), [Autos](#), [Outdoors](#)...
- [Reference](#) - [Libraries](#), [Dictionaries](#), [Phone Numbers](#)...

1996



2007



Old Days

- ▶ Browsers tolerate **Broken HTML**

```
<H1>Broken HTML</H1>
```

Hi there.

```
<p><img src=photo.png>
```

Browsers tolerate a lot of
completely broken HTML.

```
<UL>
```

```
<LI>List one
```

```
<LI>List 2
```

```
</UL>
```

World Wide Web Consortium



- Formed by Tim Berners-Lee of MIT (formerly of CERN)
- Founded to develop standards around HTML, CSS, XML, etc.
- Ensure that the web was based on open standards rather than proprietary vendor products.

<http://www.w3.org/Consortium/facts>

Following the Rules

`<h1>Broken HTML</h1>`

Hi there.

`<p>`

Browsers tolerate a lot of completely broken HTML.

`</p>`

`<ul>`

`<li>List one </li>`

`<li>List 2</li>`

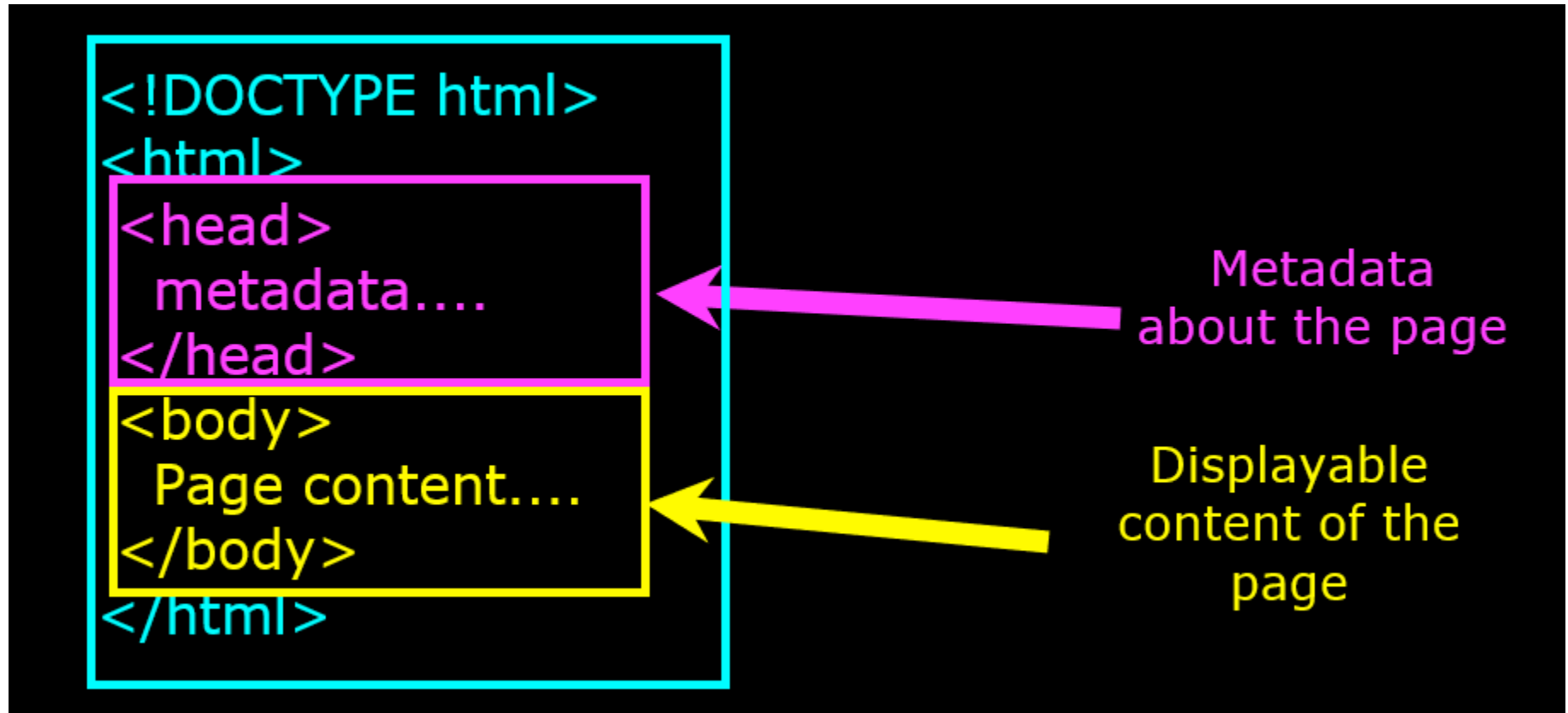
`</ul>`

Tags must be lowercase.

Attributes must be enclosed in quotes.

Tags must be closed.

Structure of an HTML Document



Multiple Files

- ▶ We can put multiple files in the same directory and then use them in relative links

 page1.html	11/2/2020 8:14 AM	Chrome HTML Do...	1 KB
 page2.html	11/2/2020 8:13 AM	Chrome HTML Do...	1 KB

HTML Tags

- ▶ Tags have a beginning and end

```
<p>You can add a style like  
<strong>bold</strong> to some text  
by enclosing it in the  
<em>appropriate</em> tag.</p>
```


HTML Tag Basics

Tags “mark up” the HTML document.
The tags are read and interpreted by
the browser but not shown.

A self-closing tag does not need a
corresponding end tag.

Start tag



`<h1>Hello World</h1>`

End tag



Attribute



``

Self-closing tag



What about special characters?

<p>Less than <**</p>**
<p>Greater than >**</p>**
<p>Ampersand &**</p>**
<p>Ampersand inception &amp;**</p>**
<p>Semicolon just works ;**</p>**
<p>Money characters: £ € ¥**</p>**
<p>Math characters: ∑ ∀ ∈**</p>**
<p>Arrows: ← ↑ → ↓**</p>**
<p>The special characters can be
in links: ****
♠♣♥&diamonds;**</p>**

There are lots of special characters

There are a few characters that we use in HTML that need a special representation to be shown on a page.

Less than <

Greater than >

Ampersand &

Ampersand inception &

Semicolon just works ;

Money characters: £ € ¥

Math characters: Σ $\forall$ $\in$

Arrows: $\leftarrow$ $\uparrow$ $\rightarrow$ $\downarrow$

The special characters can be in links: [♠♣♥♦](#)

HTML Comments

```
<!-- Ignore this for now :) -->
```

```
<p style="border-style: none; position: fixed;  
bottom: 10px; right: 10px;">Go to the  
<a href="navdetail.htm">very last page</a>.</p>
```

HTML Links

- One of the key things about HTML is making a set of pages and creating “hypertext” links amongst those pages.
- Links are what make the “web” into a web of interlinked documents.
- The interlinked nature of the web leads to the “intelligence” that search engines like Google appear to have.

```
<h1>The First Page</h1>
```

```
<p>
```

If you like, you can switch to the

```
<a href="http://www.dr-chuck.com/page2.htm">
```

Second Page.

```
</p>
```



A link is a “hot spot” on the page. It can be text or an image. Often it is visually marked to make it easier to “notice” so as to encourage users to click!

“a” is short for “anchor” and “href” is short for “hypertext reference”

```
<h1>The Second Page</h1>
```

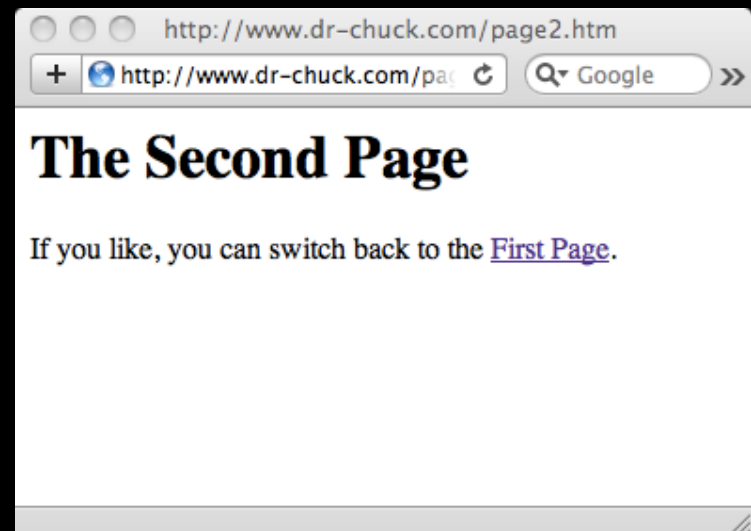
```
<p>
```

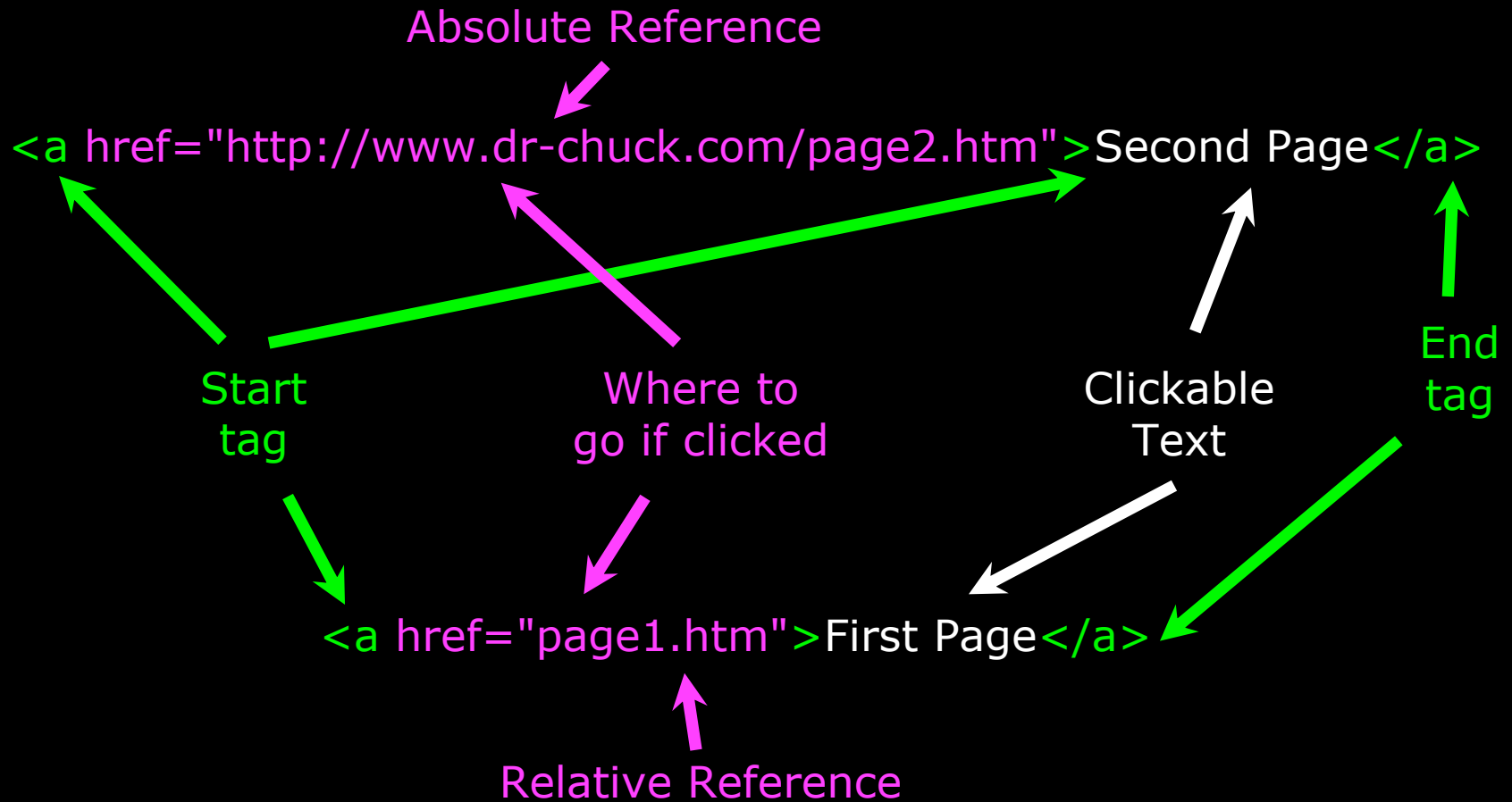
If you like, you can switch back to the

```
<a href="page1.htm">
```

```
First Page</a>.
```

```
</p>
```





Absolute vs. Relative

```
<a href="http://www.dr-chuck.com/page2.htm">Second Page</a>
```

A hypertext references can be a full URL and refer to some other page anywhere on the Internet.

```
<a href="page1.htm">First Page</a>
```

Or the reference can be a file name that is assumed to be in the same folder as the current document (relative reference).

Images

<p>

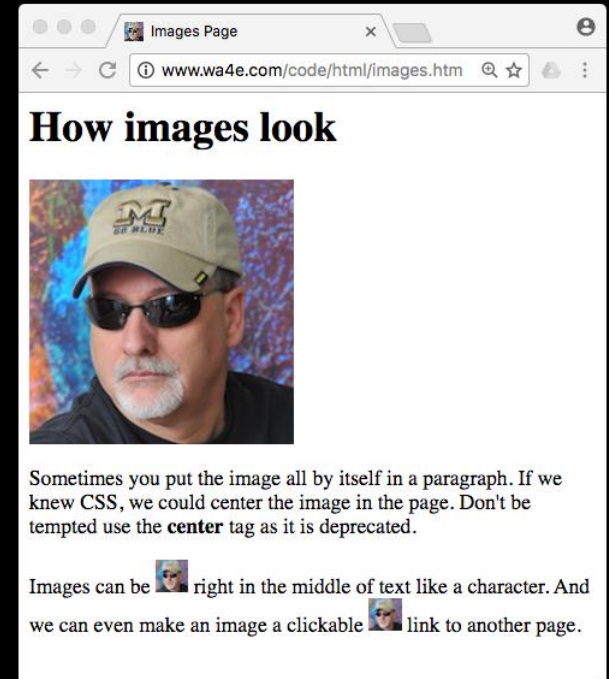
Images can be **** right in the middle of text like a character. And we can even make an image a clickable

<img

style="length:60px;width:60px"

src="tiny.png"> link to another page.

</p>



A List ...

`<ul>`

`<li><p>`This pages shows us how lists work`</p></li>`

`<li><p>`We need to encode

`<a href="special.htm">`certain characters`</a>`

to show them in HTML`</p></li>`

`<li><p>`We should learn more about

`<a href="links.htm">`Links`</a>`

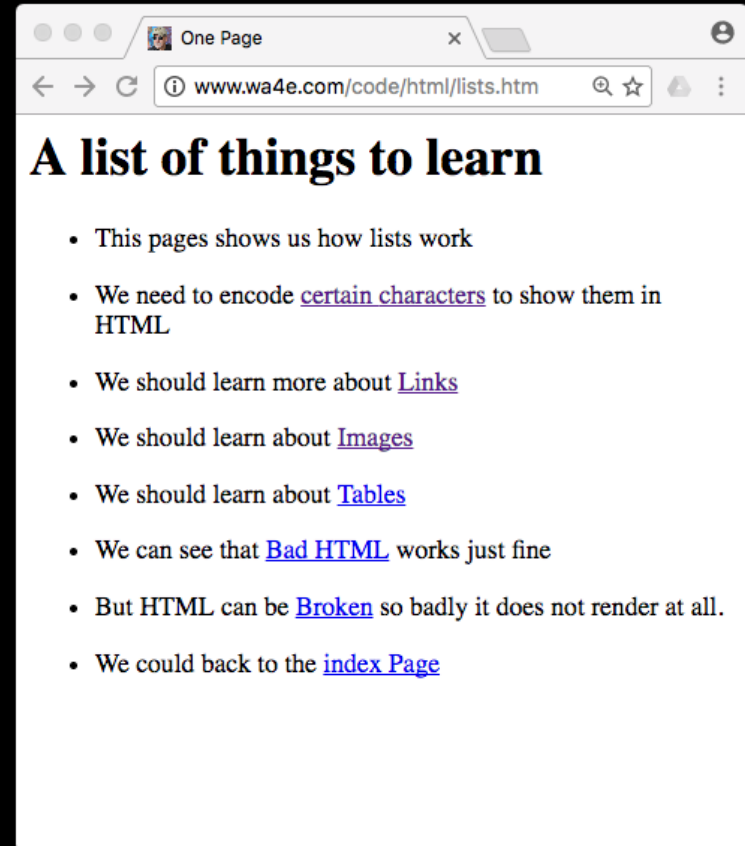
`</p></li>`

`<li><p>`We should learn about

`<a href="images.htm">`Images`</a>`

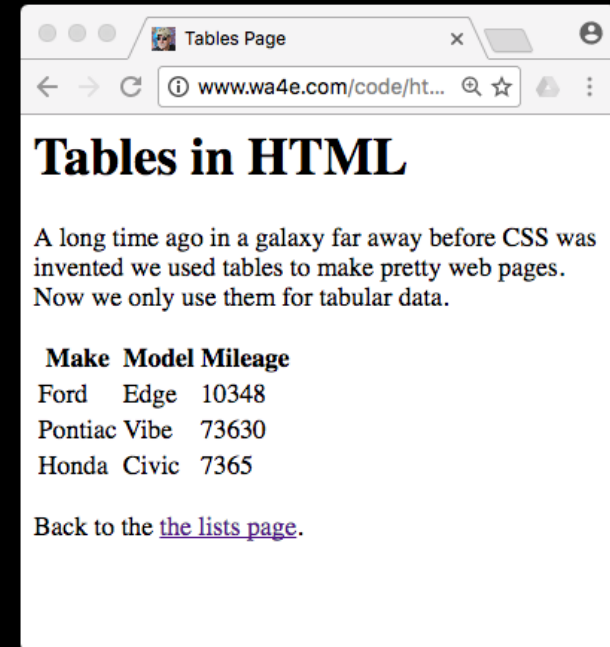
`</p></li>`

`</ul>`



Tables

```
<table>
  <tr>
    <th>Make</th>
    <th>Model</th>
    <th>Mileage</th>
  </tr>
  <tr>
    <td>Ford</td><td>Edge</td>
    <td>10348</td>
  </tr>
  <tr>
    <td>Pontiac</td><td>Vibe</td><td>73630</td>
  </tr>
</table>
```



HTML Elements - Headings

Heading 1

Heading 2

Heading 3

Heading 4

Heading 5

Heading 6

```
<h1>Heading 1</h1>
```

```
<h2>Heading 2</h2>
```

```
<h3>Heading 3</h3>
```

```
<h4>Heading 4</h4>
```

```
<h5>Heading 5</h5>
```

```
<h6>Heading 6</h6>
```

HTML Elements - Paragraphs

```
<p>
```

```
This paragraph  
contains a lot of lines  
in the source code,  
but the browser  
ignores it.
```

```
</p>
```

```
<p>
```

```
This paragraph  
contains          a lot of spaces  
in the source      code,  
but the            browser  
ignores it.
```

```
</p>
```

HTML Elements – Horizontal Rules

This is heading 1

This is some text.

This is heading 2

This is some other text.

This is heading 2

This is some other text.

```
<!DOCTYPE html>
<html>
<body>

<h1>This is heading 1</h1>
<p>This is some text.</p>
<hr>

<h2>This is heading 2</h2>
<p>This is some other text.</p>
<hr>

<h2>This is heading 2</h2>
<p>This is some other text.</p>

</body>
</html>
```

HTML Styles

I am Red

I am Blue

I am Big

```
<!DOCTYPE html>
<html>
<body>

<p>I am normal</p>
<p style="color:red;">I am red</p>
<p style="color:blue;">I am blue</p>
<p style="font-size:50px;">I am big</p>

</body>
</html>
```

HTML Styles

This is a heading

This is a paragraph.

```
<body style="background-color:powderblue;">  
  
<h1>This is a heading</h1>  
<p>This is a paragraph.</p>  
  
</body>
```


HTML Styles

```
<h1 style="font-family:verdana;">This is a heading</h1>  
<p style="font-family:courier;">This is a paragraph.</p>
```

This is a heading

This is a paragraph.

HTML Formatting

This text is bold

This text is italic

This is subscript and superscript

```
<!DOCTYPE html>
<html>
<body>

<p><b>This text is bold</b></p>
<p><i>This text is italic</i></p>
<p>This is<sub> subscript</sub> and <sup>superscript</sup></p>

</body>
</html>
|
```

HTML Formatting Elements

Formatting elements were designed to display special types of text:

- `<b>` - Bold text
- `<strong>` - Important text
- `<i>` - Italic text
- `<em>` - Emphasized text
- `<mark>` - Marked text
- `<small>` - Smaller text
- `<del>` - Deleted text
- `<ins>` - Inserted text
- `<sub>` - Subscript text
- `<sup>` - Superscript text

HTML Formatting

My favorite color is ~~blue~~ red.

```
<!DOCTYPE html>
<html>
<body>

<p>My favorite color is <del>blue</del> red.</p>

</body>
</html>
|
```

HTML Formatting

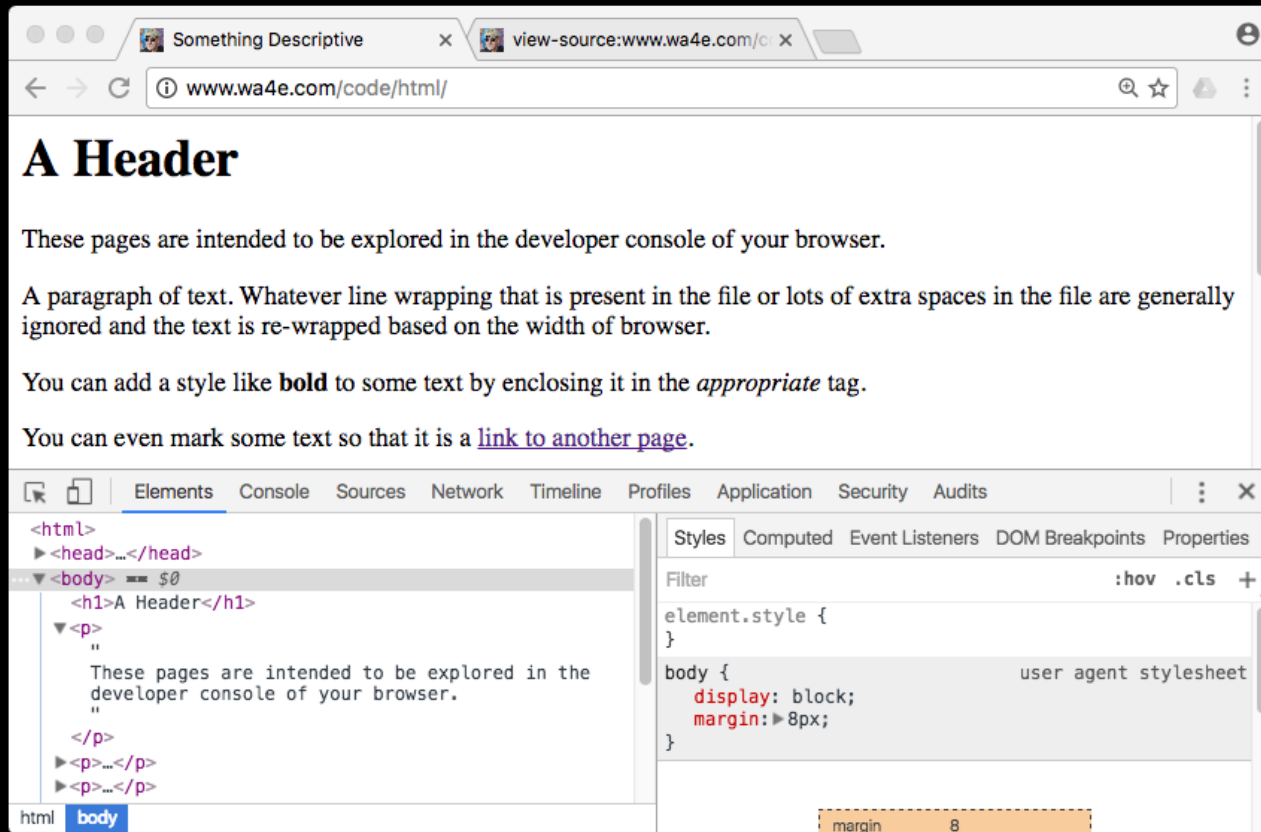
My favorite color is ~~blue~~ red.

```
<!DOCTYPE html>
<html>
<body>

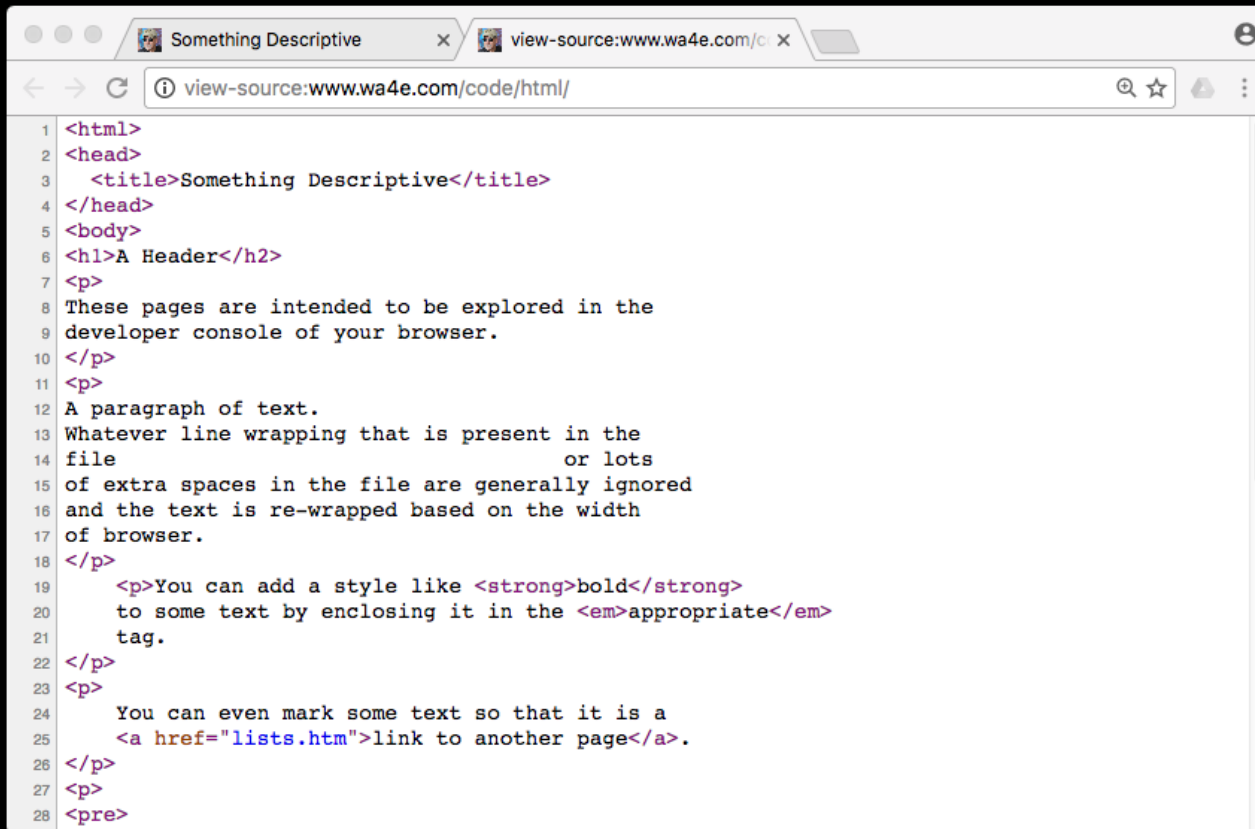
<p>My favorite color is <del>blue</del> <ins>red</ins>.</p>

</body>
</html>
```

Explore Sample Code



Explore Sample Code



The image shows a web browser window with two tabs. The active tab is titled 'view-source:www.wa4e.com/c...' and the address bar shows 'view-source:www.wa4e.com/code/html/'. The page content is HTML source code, which is displayed with line numbers from 1 to 28 on the left. The code includes a title 'Something Descriptive', a header 'A Header', and several paragraphs of text. Some text is wrapped in code tags like and to demonstrate their use. A link to 'lists.htm' is also shown.

```
1 <html>
2 <head>
3   <title>Something Descriptive</title>
4 </head>
5 <body>
6   <h1>A Header</h2>
7   <p>
8     These pages are intended to be explored in the
9     developer console of your browser.
10  </p>
11  <p>
12    A paragraph of text.
13    Whatever line wrapping that is present in the
14    file                               or lots
15    of extra spaces in the file are generally ignored
16    and the text is re-wrapped based on the width
17    of browser.
18  </p>
19    <p>You can add a style like <strong>bold</strong>
20    to some text by enclosing it in the <em>appropriate</em>
21    tag.
22  </p>
23  <p>
24    You can even mark some text so that it is a
25    <a href="lists.htm">link to another page</a>.
26  </p>
27  <p>
28  <pre>
```

HTML 5.0 New Input Fields

HTML5 introduces several new `<input>` types like email, date, time, color, range, and so on. to improve the user experience and to make the forms more interactive.

However, if a browser failed to recognize these new input types, it will treat them like a normal text box.

Source: <https://www.tutorialrepublic.com/html-tutorial/html5-new-input-types.php>

Difference: <https://www.simplilearn.com/tutorials/html-tutorial/html-vs-html5>

HTML 5.0 Web Storage

The HTML5's web storage feature lets you store some information locally on the user's computer, like cookies, but it is faster and much better than cookies.

The information stored in the web storage isn't sent to the web server as opposed to the cookies where data sent to the server with every request. Also, where cookies let you store a small amount of data (nearly 4KB), the web storage allows you to store up to 5MB of data.

HTML 5.0 Web Storage

There are two types of web storage, which differ in scope and lifetime:

Local storage — The local storage uses the `localStorage` object to store data for your entire website on a permanent basis. That means the stored local data will be available on the next day, the next week, or the next year unless you remove it.

Session storage — The session storage uses the `sessionStorage` object to store data on a temporary basis, for a single browser window or tab. The data disappears when session ends i.e. when the user closes that browser window or tab.

You can use JavaScript in order to update/retrieve/insert data inside a local storage or a session.

<https://www.tutorialrepublic.com/html-tutorial/html5-web-storage.php>

HTML5 Application Cache

Typically most web-based applications will work only if you're online. But HTML5 introduces an application cache mechanism that allows the browser to automatically save the HTML file and all the other resources that needs to display it properly on the local machine, so that the browser can still access the web page and its resources without an internet connection.

HTML5 Application Cache

Advantages of HTML5 application cache feature:

Offline browsing — Users can use the application even when they're offline or there are unexpected disruptions in the network connection.

Improve performance — Cached resources load directly from the user's machine rather than the remote server hence web pages load faster and performing better.

Reduce HTTP request and server load — The browser will only have to download the updated/changed resources from the remote server that minimize the HTTP requests and saves precious bandwidth as well as reduce the load on the web server.

Summary

- HTML has gone through many changes and evolutions.

It started clean and simple - then got ugly and nasty - now we are back to a clean and simple approach.

- A key to modern/clean HTML is that formatting is handled using CSS - Cascading Style Sheets.